

Random Variables and Limit Theorems

Course notes - Department of Mathematics

1. Probability spaces

A probability space consists of a sample space of outcomes, a sigma-algebra of measurable events, and a measure assigning each event a number in $[0,1]$ with the whole space having measure one and countable additivity holding for disjoint events.

Conditional probability $P(A|B) = P(A \text{ and } B)/P(B)$ renormalises to the information that B occurred. Independence means $P(A \text{ and } B) = P(A)P(B)$, which is a statement about the measure, not about causation.

2. Random variables

A random variable is a measurable function from the sample space to the reals. Its distribution is characterised by the cumulative distribution function, and where it exists by a probability mass or density function.

Expectation is the integral of the variable against the measure and is linear regardless of dependence. Variance measures spread and adds only for uncorrelated variables.

3. Standard distributions

The Bernoulli distribution models a single trial, and the binomial counts successes across independent trials. The Poisson distribution arises as the limit of rare events over a fixed interval and models arrival counts.

The normal distribution is characterised by its mean and variance and is the limit that the central limit theorem produces. The exponential distribution models waiting times and is the unique continuous memoryless distribution.

4. Bayes theorem

Bayes theorem inverts a conditional: $P(A|B) = P(B|A)P(A)/P(B)$. It is the formal mechanism for updating a prior belief in light of evidence to yield a posterior.

The base rate fallacy is what happens when the prior is ignored. A test that is 99 percent accurate for a condition affecting one person in ten thousand still yields mostly false positives, because the prior dominates.

5. Limit theorems

The law of large numbers states that the sample mean of independent identically distributed variables converges to the expectation, in probability for the weak form and almost surely for the strong form.

The central limit theorem states that the standardised sample mean converges in distribution to the standard normal, whatever the underlying distribution, provided the variance is finite. This is why the normal distribution appears so widely and why it underpins confidence intervals.